Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor | Draft for review
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 6 August 2026 Time: Not recorded
Group members present	To be confirmed
Apologies	To be confirmed

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Review relevant literature on rheological signal analysis, Fourier methods and wavelets.	All members	To be confirmed
Inspect available blood rheology data and identify measurement protocols.	All members	To be confirmed

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Literature and implementation direction	Proposed focus: move from reviewing papers towards implementing selected preprocessing, feature extraction and evaluation techniques in Python notebooks.
Data understanding	Review the available Excel recordings and distinguish LAOS, frequency sweep and step protocols. Identify time, stress, strain and shear-rate channels before comparing methods.
Consistent preprocessing	Consider a common loading and preprocessing workflow so that methods can be evaluated on comparable signals. Check timestamps, sampling rates, missing values and signal quality before choosing resampling settings.
Candidate signal processing methods	Consider Fourier methods, including FFT and STFT, alongside continuous and discrete wavelet transforms. Compare their ability to summarize periodic responses and localize changes over time.
Physical features and validation	Consider spectral entropy and harmonic distortion, together with rheological transition indicators. Clarify how an estimated change relates to gelation before interpreting it as a clotting time.
Benchmarking and next steps	Explore a controlled synthetic signal with a known transition for method evaluation, alongside real recordings for practical relevance. Prepare an initial notebook demonstration and questions for supervisor feedback.

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3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Prepare an initial notebook covering data loading, protocol classification and signal quality checks.	Rohan	For review on 13 August
Implement an initial Fourier and wavelet comparison on common signals.	Rohan	For review on 13 August
Investigate synthetic benchmarking with a known transition and nonlinear harmonics.	Rohan	For review on 13 August
Continue literature review and identify techniques suitable for implementation.	All members	For review on 13 August
Prepare questions on physical interpretation, model evaluation and limitations of the available data.	All members	For review on 13 August

4 Next meeting

Location	Not recorded
Date & Time	13 August 2026; time to be confirmed
Goals	Review the initial analysis pipeline, candidate features, validation strategy and implementation progress.

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